The second law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain the direction of heat transfer and the limits of energy conversion.

SUCCESS CRITERIA - I can...

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **0.3 kg** metal cools from **91** to **31°C** while **0.25 kg** water warms from **11** to **31°C**.
1. Use water c=**4200**; neglect cup/losses.
2. **Find metal c**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

Worked reasoning: Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

Your turn: Draw before-and-after temperature states and the spontaneous heat-flow direction.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A **50°C** block touches a **10°C** block. **Draw net heat flow** and **predict their trend toward equilibrium**.

50°C

10°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

During one modeled refrigerator cycle, **200 J** is removed from the cold compartment using **50 J** of electrical work.
1. **Draw the energy flows**.
2. **Find the energy released** to the room and the coefficient of performance, Q_c/W.
3. **Explain why COP** is not heat-engine efficiency.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **0.4 kg** object moving at **6 m/s** stops through friction over **0.8 m**. **Find energy transformed** and average opposing-force magnitude.
Show the energy changes and transfers; **label units**.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Two bodies exchange heat in an isolated system and reach the same temperature. Is the total energy lost? **What happens** to the ability to extract work from their temperature difference?

Sample A Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A refrigerator removes **200 J** from a cold space while receiving **80 J** electrical work. **Find heat delivered** to the room and **explain why it** is larger than **200 J**.

Sample A Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3